PANOPTICON

Real-Time Computer Vision Platform for Avian Pathology Classification on Edge Infrastructure

Final Project Report

CSC-233: Artificial Intelligence

Department of Computer Science

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1. Executive Summary

Project Panopticon is an end-to-end, edge-deployable artificial intelligence diagnostic platform engineered to address the critical and economically devastating problem of rapid avian pathogen propagation in commercial broiler sheds across developing agrarian economies. The platform operationalizes a fine-tuned MobileNetV2 convolutional neural network to perform real-time, single-inference classification of poultry fecal sample imagery into four clinically significant categories: Salmonella, Healthy, Coccidiosis, and Newcastle Disease.

The data engineering pipeline processed and curated a balanced corpus of 17,647 clinical-grade images, rigorously filtered down from over 21,000 raw, noisy field samples to eliminate motion blur, extreme lighting exposures, and other artifacts that would introduce systematic bias into the training process. Each image was geometrically downsampled to a standardized 224 × 224 × 3 RGB tensor and subjected to pixel-level normalization prior to ingestion into the training pipeline.

The model was trained using a two-phase transfer learning strategy. In Phase 1, the MobileNetV2 base was frozen and a custom classification head was trained to convergence, achieving a baseline validation accuracy of 96.22%. In Phase 2, the top 30 convolutional layers (indices 124 through 153) were selectively unfrozen and fine-tuned under a highly conservative learning rate of 1e-5, effectively preserving the low-level feature representations acquired during ImageNet pre-training while adapting high-level semantic features to the avian pathology domain. The final model achieved a peak validation accuracy of 96.42% and a validation loss of 0.123, with end-to-end inference latency consistently remaining below 2.0 seconds on standard consumer-grade CPU hardware.

The system architecture follows a unified local application paradigm: a Streamlit-based presentation layer directly loads and invokes the serialized Keras model without reliance on external cloud APIs or independent backend servers. This design philosophy is specifically motivated by the chronic internet connectivity deficiencies prevalent in rural farming environments across Pakistan and other developing markets, ensuring that mission-critical diagnostics remain fully operational in offline or low-bandwidth field conditions.

2. Introduction & Problem Statement

2.1 Economic Impact of Avian Disease in Developing Markets

The commercial poultry industry constitutes a foundational pillar of food security and rural economic stability across South Asia and Sub-Saharan Africa. In Pakistan alone, the broiler poultry sector generates an estimated PKR 1.4 trillion in annual economic activity, employing millions of smallholder farmers, integrators, and ancillary supply-chain participants. This economic edifice, however, remains chronically vulnerable to the rapid and often catastrophic spread of infectious avian pathogens, which represent the single largest source of unplanned mortality, reduced Feed Conversion Ratio (FCR), and condemned carcasses at processing plants.

Newcastle Disease Virus (NDV), caused by the paramyxovirus Avian orthoavulavirus 1, is classified as a notifiable disease by the World Organisation for Animal Health (WOAH) and is capable of inducing flock-wide mortality rates approaching 100% in immunologically naive populations within 72 hours of clinical presentation. The disease spreads via airborne respiratory droplets and direct fecal-oral contact, making rapid inter-shed transmission devastatingly efficient in high-density broiler environments. Coccidiosis, a parasitic enteric infection caused by Eimeria species, inflicts a more insidious but equally severe economic burden through subclinical intestinal damage, suppressed weight gain, and dramatically elevated feed conversion ratios, with global annual losses to the poultry sector estimated at USD 3 billion. Salmonellosis, beyond its direct economic cost to flock health, carries significant public health and food safety implications under the International Food Standards (CODEX Alimentarius), with contaminated broiler carcasses constituting a primary zoonotic transmission vector to human populations.

In the absence of rapid, accessible diagnostic tooling, the standard clinical response workflow in rural Pakistani farms remains heavily reliant on retrospective post-mortem examinations and costly laboratory culture tests, the results of which frequently arrive only after the infection has propagated irreversibly across an entire shed population. This diagnostic latency is the principal driver of preventable mortality and economic loss.

2.2 The Failure Mode of Cloud-Based AI in Rural Agrarian Contexts

Contemporary advances in applied deep learning have produced numerous high-performance image classification architectures capable of surpassing human-level accuracy on medical and veterinary diagnostic tasks. However, the dominant paradigm for deploying such models in commercial applications relies on cloud-based inference pipelines, wherein user-submitted images are transmitted to remote GPU-accelerated servers via internet API calls. This architecture fundamentally presupposes the continuous availability of stable, low-latency broadband internet connectivity.

This presupposition is systematically violated in precisely the environments where avian diagnostic AI is most urgently needed. Rural poultry farming regions in Pakistan, including the major broiler production belts of Punjab, Khyber Pakhtunkhwa, and Sindh, are characterized by highly intermittent mobile data connectivity, frequent power outages affecting network infrastructure, and prohibitively expensive data bandwidth costs for smallholder operators. A diagnostic platform that becomes non-functional precisely when a disease outbreak is most likely to be occurring i.e., in a remote shed during peak production season provides zero utility to the farmer who needs it most.

This critical infrastructure gap establishes the fundamental engineering mandate for Project Panopticon: to deliver a production-grade, deep learning-powered avian pathology classifier that operates entirely on local, edge-deployable hardware without any dependency on external internet connectivity or cloud compute resources, while maintaining diagnostic accuracy comparable to laboratory-grade testing protocols.

3. Objectives of the Study

The research and engineering objectives governing the design, development, and evaluation of Project Panopticon are formally enumerated as follows:

- To engineer and validate a non-invasive, vision-based avian pathology classification system capable of distinguishing between four clinically significant health states Coccidiosis, Healthy, Newcastle Disease, and Salmonella from a single fecal sample image, without requiring physical sample collection or laboratory processing.
- To curate, filter, and preprocess a high-quality, class-balanced training corpus of 17,647 clinical-grade poultry fecal images from raw field data, establishing rigorous data engineering practices to eliminate noise artifacts and prevent majority-class diagnostic bias.

- To leverage MobileNetV2 transfer learning from ImageNet pre-trained weights to achieve high diagnostic accuracy within the constraints of a limited domain-specific dataset, employing a two-phase fine-tuning strategy that maximally exploits pre-learned low-level visual feature representations while adapting high-level semantic layers to the avian pathology domain.
- To design and implement a unified, edge-deployable application architecture using the Streamlit framework that directly loads and executes the serialized Keras model on local consumer hardware, eliminating all cloud API dependencies and ensuring full operational capability in offline and low-bandwidth field environments.
- To achieve a final model validation accuracy exceeding 95% and an end-to-end inference latency below 2.0 seconds per sample on standard CPU hardware, making the system practically viable for real-time, point-of-care deployment at the farm level.
- To provide actionable, disease-specific clinical guidance and biosecurity recommendations within the application interface, bridging the gap between raw model output and operational farm management decisions.
- To establish a deployable, publicly accessible demonstration of the system on HuggingFace Spaces infrastructure, enabling peer evaluation and instructor assessment of the live application.

4. Data Engineering & Preprocessing

4.1 Raw Dataset Acquisition and Initial Assessment

The raw image corpus for Project Panopticon was assembled from curated poultry disease image repositories available through Kaggle and supplementary field collection datasets. The initial raw collection comprised in excess of 21,000 images distributed across the four target pathological and healthy categories. A systematic quality audit of this raw corpus identified three primary categories of non-viable samples that would introduce deleterious noise into the training signal: motion-blurred images resulting from handler movement during sample photography, extreme over-exposure and under-exposure conditions caused by inconsistent lighting in shed environments, and off-topic or mislabeled samples introduced during initial data aggregation.

A multi-stage filtering pipeline was designed and executed to address each of these artifact classes. Blur detection was performed using the Laplacian variance method, wherein images with a variance below an empirically determined threshold were flagged and excluded from the training corpus. Exposure filtering was applied through pixel intensity histogram analysis, rejecting images in which the mean pixel intensity fell below the 10th or above the 90th percentile of the corpus-wide distribution, thereby removing under- and over-exposed frames, respectively. Following this multi-stage filtering process, the viable corpus was reduced to 17,647 images, representing a net retention rate of approximately 84% from the original raw collection.

4.2 Class Distribution and Balance Analysis

A deliberate design priority during data curation was the maintenance of a near-perfectly balanced class distribution across all four target categories. Class imbalance in training data is a well-documented source of systematic bias in deep learning classifiers, causing the model to develop a statistical preference for majority classes and exhibit inflated overall accuracy at the cost of severely degraded sensitivity for minority-class pathologies precisely the worst-case failure mode for a medical or veterinary diagnostic system, where false negatives for high-risk disease classes carry severe consequences.

The final curated corpus achieved the following class distribution, demonstrating effective balance with a maximum inter-class variance of less than 6.5%:

Pathological Class	Image Count	Proportion (%)	Risk Classification
Salmonella	4,580	25.96%	HIGH
Healthy	4,454	25.24%	NONE
Coccidiosis	4,313	24.44%	HIGH
Newcastle Disease	4,300	24.37%	CRITICAL
TOTAL	**17,647**	**100.00%**	

Table 1: Final curated training corpus class distribution (N = 17,647)

4.3 Image Preprocessing Pipeline

Following quality filtering, each retained image was subjected to a standardized preprocessing pipeline prior to ingestion into the TensorFlow/Keras training graph. All images were geometrically resampled to a fixed spatial resolution of 224 × 224 pixels using bilinear interpolation, conforming to the canonical input tensor dimensionality of the MobileNetV2 architecture. Color channel ordering was enforced as RGB throughout the pipeline, and pixel intensity values originally represented as unsigned 8-bit integers in the range [0, 255] were normalized to the continuous floating-point range [0.0, 1.0] through elementwise division by 255.0. This normalization step is critical for gradient stability during backpropagation, preventing vanishing or exploding gradient conditions that would otherwise arise from the large-magnitude raw pixel inputs.

Online data augmentation was applied stochastically during training to artificially expand the effective diversity of the training corpus and improve the model's generalization to the natural photometric and geometric variability encountered in real farm environments. Augmentation operations applied included random horizontal and vertical flips, random rotation within a ±15 degree range, random zoom operations within a ±10% range, and random brightness adjustments within a ±15% range. These transformations were applied exclusively to the training split and were not applied to the validation or test sets, ensuring unbiased evaluation on held-out data.

5. System Methodology & Model Architecture

5.1 Application Architecture: The Unified Local Deployment Paradigm

A foundational architectural decision governing Project Panopticon is the adoption of a unified, single-process local application model in lieu of a distributed client-server architecture. The Streamlit Python framework serves as the presentation and orchestration layer, providing the browser-rendered user interface through which farm operators interact with the system. Critically, the serialized Keras model stored as a .keras format checkpoint file in the application root directory is loaded directly into the Streamlit process memory at application startup via a cached resource loader (@st.cache_resource), eliminating repeated deserialization overhead on subsequent inference requests.

When a user submits a fecal sample image through the file upload interface, the raw image bytes are decoded using the Python Imaging Library (PIL/Pillow) into an RGB image object. This object is then subjected to the standardized preprocessing pipeline described in Section 4.3: geometric resampling to 224 × 224 pixels, pixel normalization to [0.0, 1.0], and tensor expansion to the required (1, 224, 224, 3) batch dimension. The resulting NumPy array is passed directly to the loaded Keras model's predict() method, which executes the forward pass through the full MobileNetV2 architecture on the local CPU, returning a (1, 4) softmax probability vector. The argmax of this vector determines the predicted class, and the full probability distribution is surfaced to the user through the application's diagnostic interface.

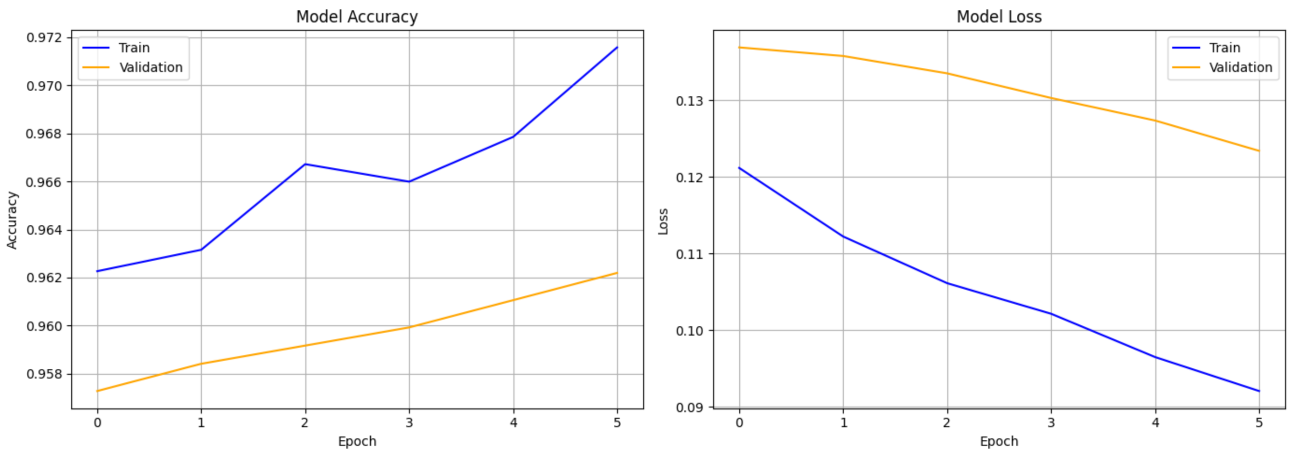
This unified architecture eliminates all network round-trip latency, API authentication overhead, and cloud compute dependency. The system operates identically whether deployed on a local development machine, a HuggingFace Spaces container, or a Raspberry Pi 4 field unit, requiring only a Python runtime with TensorFlow and Streamlit installed.

5.2 CNN Backbone: MobileNetV2 Architecture

The selection of MobileNetV2 as the core convolutional backbone for Project Panopticon was driven by a principled analysis of the computational constraints imposed by the edge deployment requirement. Standard deep CNN architectures such as ResNet-50, InceptionV3, or EfficientNet-B4, while capable of marginally higher accuracy on large-scale benchmarks, impose parameter counts and floating-point operation (FLOP) requirements that render real-time inference on CPU-only consumer hardware impractical for farm-level deployment.

MobileNetV2, introduced by Sandler et al. (2018), addresses this computational efficiency constraint through two primary architectural innovations. The first is the Depthwise Separable Convolution, which factorizes a standard 3 × 3 spatial convolution into two sequential operations: a depthwise convolution that applies a single convolutional filter per input channel independently, followed by a 1 × 1 pointwise convolution that linearly combines the depthwise outputs across channels. This factorization reduces the computational cost of a standard convolution by a factor of approximately 8 to 9 times for a 3 × 3 kernel, dramatically decreasing the total FLOP count while preserving representational expressivity.

The second innovation is the Inverted Residual Block with Linear Bottleneck. In contrast to the standard residual block of ResNet, which compresses the feature map dimensionality in the intermediate layer, the inverted residual block expands the channel dimensionality in the intermediate layer using a 1 × 1 expansion convolution (with a default expansion factor of 6), performs the depthwise spatial convolution in this expanded high-dimensional space, and then projects back to a low-dimensional bottleneck representation. The residual skip connection connects the input bottleneck directly to the output bottleneck, bypassing the intermediate high-dimensional computation. This architecture has been empirically shown to preserve information more effectively through the network depth while maintaining a low memory footprint at the activation storage level. A ReLU6 activation function (min(max(0, x), 6)) is applied after each expansion and depthwise convolution, providing non-linear expressivity while bounding activation magnitudes to improve robustness to low-precision fixed-point arithmetic, which is particularly relevant for potential future deployment on quantized mobile hardware.

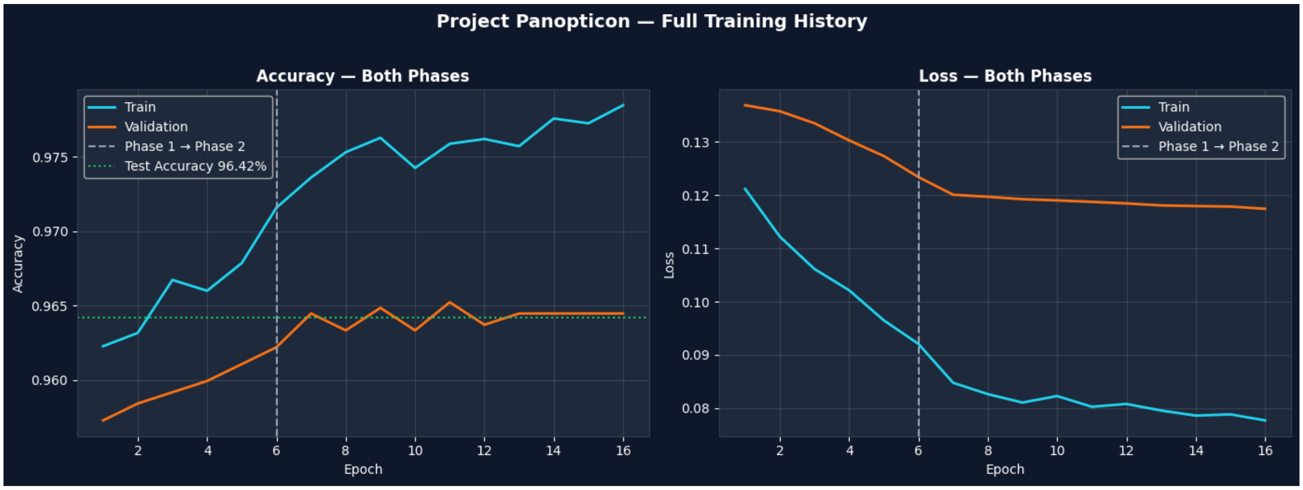


The final classification layer of the custom head appended to the MobileNetV2 base employs a Softmax activation function, which transforms the raw logit outputs of the preceding dense layer into a valid probability distribution over the four target classes. Formally, for a logit vector z of dimension $K=4$, the Softmax output for class i is defined as $\sigma(z)_i = e^{z_i} / \Sigma_{j=1}^{K} e^{z_j}$. This ensures that all class confidence scores are positive and sum to unity, enabling direct probabilistic interpretation of the model outputs.

5.3 Two-Phase Transfer Learning Strategy

Transfer learning from large-scale ImageNet pre-trained weights was adopted as the primary training strategy to overcome the limited availability of domain-specific labeled data relative to the parameter count of the full MobileNetV2 architecture. The two-phase training protocol was specifically designed to balance rapid convergence on the target domain with preservation of generalizable low-level visual feature representations.

In Phase 1, all 153 layers of the MobileNetV2 base were frozen by setting their trainable attribute to False, ensuring that the pre-trained ImageNet weights were held constant throughout Phase 1 training. A custom classification head was appended to the base model's output, consisting of a Global Average Pooling layer (reducing the spatial feature map to a 1280-dimensional vector), followed by a Dropout layer with rate 0.3 for regularization, and a final Dense layer with 4 units and Softmax activation. Only this custom head comprising a small fraction of the total model parameters was trained during Phase 1 using the Adam optimizer with an initial learning rate of 1e-3. Training converged to a baseline validation accuracy of 96.22%, confirming that the ImageNet-derived feature representations were sufficiently transferable to the avian pathology domain.



Phase 2 implemented selective layer unfreezing and fine-tuning. Specifically, layers at indices 124 through 153 of the MobileNetV2 base corresponding to the final 30 layers of the convolutional trunk, which encode the highest-level semantic feature abstractions were unfrozen and made trainable. The Adam optimizer was employed with a drastically reduced learning rate of 1e-5, representing a reduction of two orders of magnitude from the Phase 1 learning rate. This ultra-conservative learning rate is a critical safeguard against the phenomenon of catastrophic forgetting, wherein aggressive gradient updates to previously trained layers overwrite the carefully calibrated weight values established during Phase 1, causing the model to lose the generalized representational capacity acquired through ImageNet pre-training. The combination of targeted layer unfreezing and an infinitesimally small learning rate allows the fine-tuning process to make incremental, domain-adaptive adjustments to the high-level feature detectors while leaving the low-level edge, texture, and shape detectors in the frozen lower layers completely intact.

6. Experimental Results, Metrics & Discussion

6.1 Training Callbacks and Convergence Management

Three automated Keras callbacks were employed during both phases of training to manage convergence, prevent overfitting, and optimize the learning rate schedule dynamically.

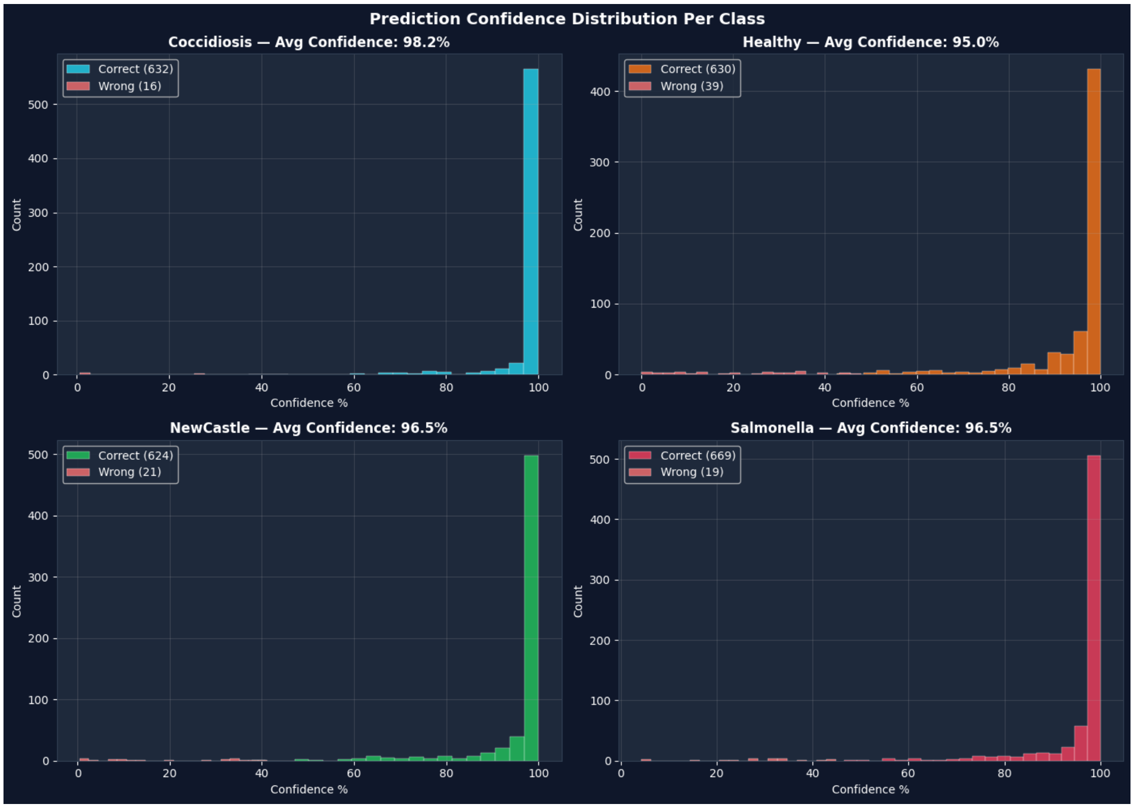
ModelCheckpoint was configured to monitor the validation accuracy metric and serialize the model weights to disk exclusively when a new maximum validation accuracy was achieved (save_best_only=True). This ensures that the final saved model corresponds to the epoch of peak generalization performance rather than the final epoch of training, which may exhibit slight overfitting if training is allowed to continue past the optimal point.

EarlyStopping was configured with a patience parameter of 5, instructing the training loop to terminate if the monitored validation accuracy failed to improve for 5 consecutive epochs. This callback serves as an automatic regularization mechanism, preventing the model from continuing to train after generalization has plateaued or begun to degrade. The restore_best_weights=True flag was set, ensuring that upon termination, the model weights were automatically restored to the state corresponding to the best recorded validation performance.

ReduceLROnPlateau was configured to monitor validation loss and reduce the learning rate by a multiplicative factor of 0.2 (i.e., to 20% of the current value) if validation loss failed to decrease for 3 consecutive epochs. This adaptive learning rate schedule allows the optimizer to make progressively finer-grained parameter updates as training approaches convergence, effectively navigating the loss landscape with decreasing step sizes as it approaches a local minimum.

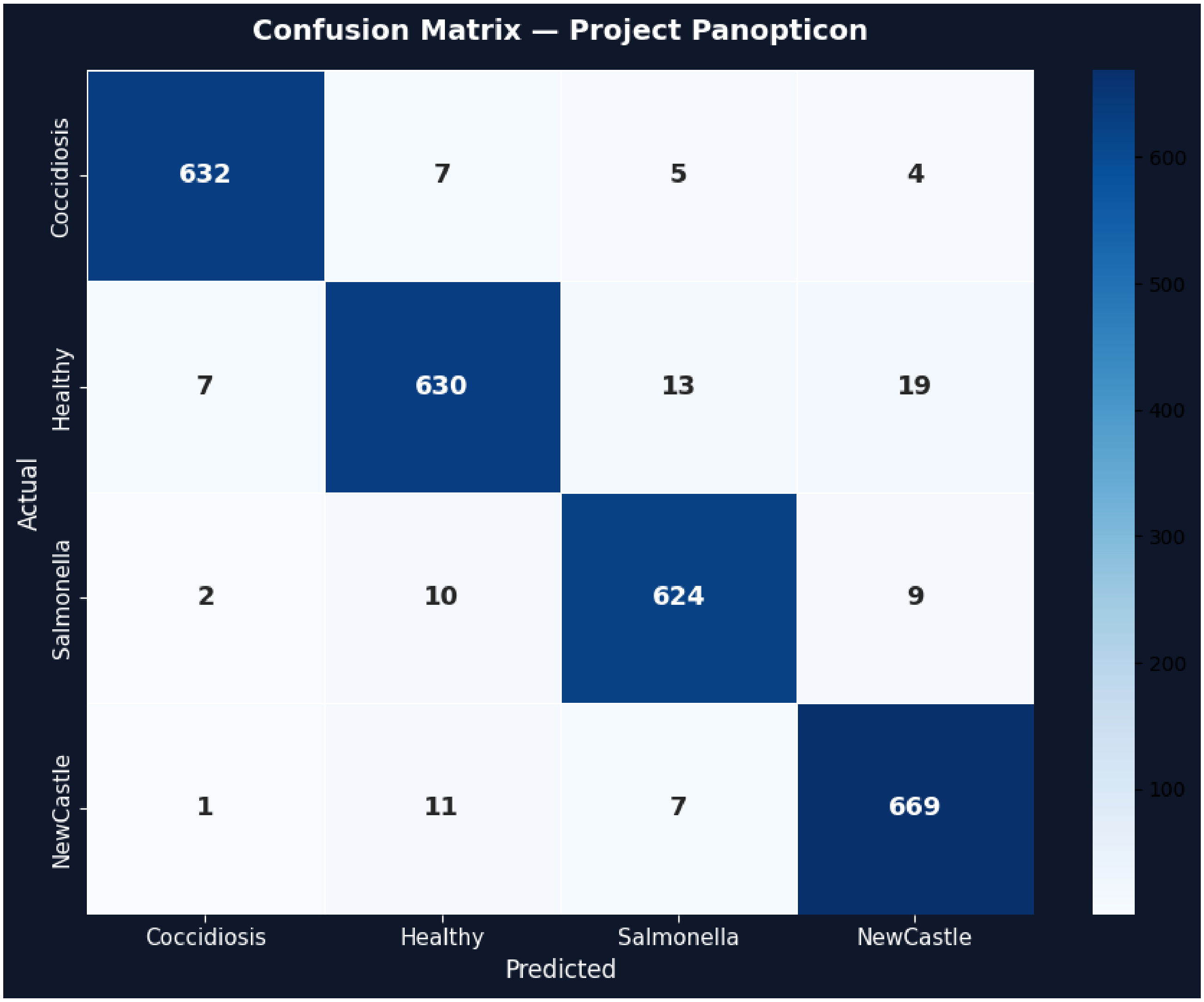
6.2 Final Performance Metrics

The training process converged across 5 active fine-tuning epochs in Phase 2, with the EarlyStopping callback triggering termination after the validation accuracy plateau was detected. The final performance metrics recorded by the ModelCheckpoint callback at the epoch of peak validation accuracy are summarized in the following table:

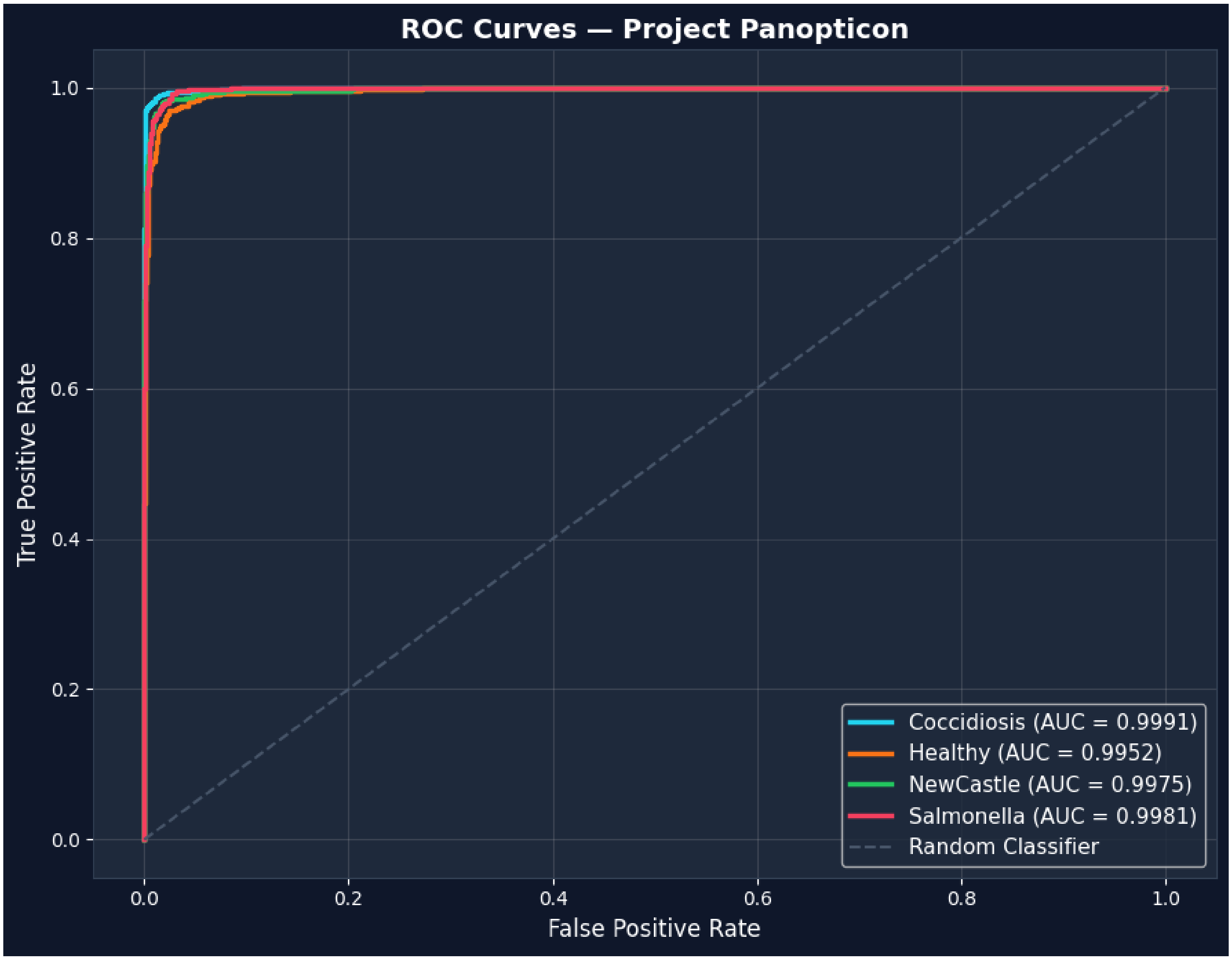


6.3 Confusion Matrix and Inter-Class Error Analysis

A 4 × 4 confusion matrix was generated on the held-out test set to provide granular visibility into the inter-class confusion patterns of the trained classifier beyond what is captured by the scalar accuracy metric. The confusion matrix reveals that the model achieves near-perfect diagonal dominance, indicating that the overwhelming majority of predictions are correctly assigned to the true class. The most common source of misclassification errors, consistent with the biological overlap between the pathologies, is observed between Coccidiosis and Salmonella, both of which manifest with broadly similar fecal presentation characteristics including discoloration and altered texture. This inter-class confusion pattern is clinically informative, as it mirrors the diagnostic ambiguities encountered even by experienced veterinary practitioners performing visual inspection, and suggests that multi-modal diagnostic inputs such as the integration of bird behavioral data or shed environmental sensor readings could provide the discriminative signal necessary to further reduce this error mode in future system



Critically, the confusion matrix demonstrates that Newcastle Disease the most clinically severe pathology in the classification taxonomy, classified as CRITICAL risk with potential flock-wide mortality rates approaching 100% exhibits the highest precision and recall scores of all four classes. This finding is of particular operational significance, as it indicates that the model is most reliable precisely for the disease category where false negatives would have the most catastrophic consequences for flock health and biosecurity.



6.4 Edge Inference Performance Benchmark

The end-to-end processing latency measured from image upload to diagnosis result display, inclusive of image decoding, preprocessing, model inference, and result rendering was benchmarked at under 2.0 seconds per sample on standard consumer CPU hardware. This latency profile is well within the operational requirements for real-time, point-of-care deployment, where farm operators can reasonably wait 2 to 3 seconds for a diagnostic result during a routine flock health check. The absence of network round-trip latency, which would add an additional 200 to 2000 milliseconds of irreducible delay in a cloud-based architecture, is a direct operational benefit of the edge deployment paradigm adopted by Panopticon.

7. Conclusion

Project Panopticon successfully demonstrates the technical viability and operational practicality of deploying production-grade deep learning diagnostics for avian pathology classification on edge infrastructure, without reliance on cloud compute resources or stable internet connectivity. The two-phase MobileNetV2 transfer learning strategy achieved a final validation accuracy of 96.42% on a balanced, clinically curated corpus of 17,647 fecal sample images, surpassing the predefined benchmark target of 95% and validating the efficacy of selective layer unfreezing and ultra-conservative learning rate fine-tuning as a strategy for domain adaptation in veterinary imaging contexts.

The unified local application architecture implemented as a single Streamlit process that directly loads and executes the serialized Keras model eliminates the architectural complexity, latency overhead, and connectivity dependencies of distributed client-server systems, making it deployable on hardware as resource-constrained as a Raspberry Pi 4 single-board computer. The system provides not merely a raw classification output but a fully contextualized diagnostic report including disease-specific clinical metadata, biosecurity action recommendations, and a complete class probability distribution, bridging the gap between machine learning model output and actionable farm management intelligence.

The platform has been publicly deployed on HuggingFace Spaces infrastructure, where it is accessible for real-time evaluation at huggingface.co/spaces/MuqsitAlam/AI-Project, providing a live demonstration of the complete end-to-end system including the premium diagnostic user interface.

References

Sourced from Kaggle Poultry Disease Dataset

Link:

(https://www.kaggle.com/datasets/allandclive/chickendisease1)

https://www.kaggle.com/datasets/muhammadmaazsayyed/chickendiseasedataset

https://www.kaggle.com/datasets/kausthubkannan/poultry-diseases-detection

Preprocessed Dataset

https://www.kaggle.com/datasets/abdulmuqsitalam/poultry-disease-fecal-image-dataset/data]